2022B F=ma Exam: Problem 11

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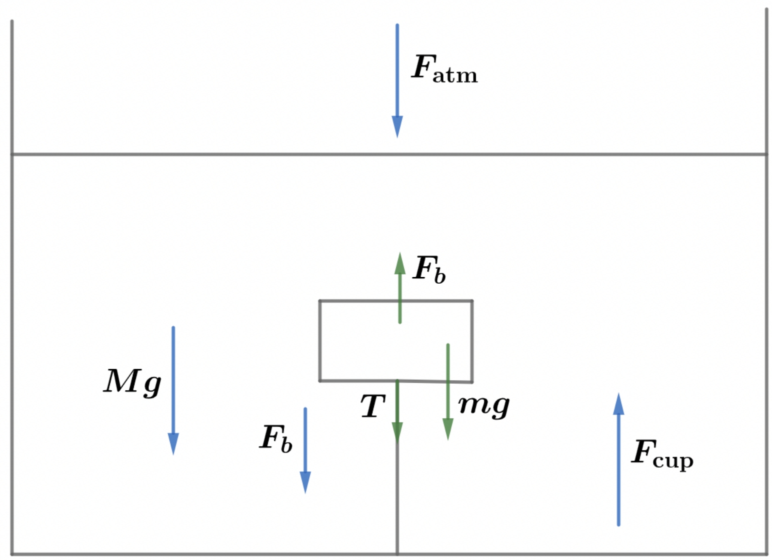

Balancing forces on the water, we have

$$F_{\text{cup}} = Mg + F_{\text{atm}} + F_b$$

where F_{cup} is the force on the water from the bottom of the cup, F_{atm} is the force on the water from the air, and F_b is the reaction force to the buoyant force on the block. We have

$$F_{\text{atm}} = P_{\text{atm}} A$$

The buoyant force is

$$F_b = \rho V_{\text{block}} g = \rho \left(\frac{m}{\rho/2} \right) g = 2mg$$

Thus,

$$F_{\text{cup}} = Mg + P_{\text{atm}} A + 2mg$$

so the answer is $\boxed{\text{E}}$.